

LEANR Diet Platform

Product Requirements Document (PRD) · v2.0 · 14 July 2026



Product	LEANR Diet Platform — AI-assisted clinical diet counselling & weekly plan management for dietitians
Owner	LEANR by Fitelo (rayan.ranaut@fitelo.co)
Status	v2.0 shipped — live at github.com/sahajverma-arch/dietplanner
Platforms	Web (desktop-first, responsive) · Vercel + Supabase + NVIDIA NIM + QStash
v2.0 highlights	LeanR Premium counselling (Q1–Q105) · AI independent clinical review · INDB/USDA nutrition grounding · Google Workspace login · ops-sheet sync (Today's Activity) · admin module

1. OVERVIEW

Dietitians run a 55–65 minute first-counselling call where they collect a client's goals, medical history, actual food day, preferences, training and behaviour patterns — then traditionally spend significant additional time manually composing a personalised 1-week diet chart, formatting it, and repeating a lighter version weekly per client. The LEANR Diet Platform removes the composition and formatting work while *raising* the clinical bar: the dietitian fills the structured **LeanR Premium counselling** live during the call (option-select, conversational — options are not read out), records their professional strategy hypothesis, and the platform's AI **independently reviews** that hypothesis against the complete client profile before generating a restriction-aware, macro-annotated 7-day plan whose numbers are grounded in the ICMR-NIN/INDB and USDA food databases — delivered as a branded, client-ready PDF in about a minute. Weekly follow-ups take a short check-in and produce the next adjusted week.

2. GOALS & NON-GOALS

Goals

- Cut per-client plan preparation time from hours to minutes while keeping the dietitian in clinical control.
- Run the full **LeanR Premium Final New Client First Counselling** (Q1–Q105, 12 sections) as a live, schema-driven form with conditional follow-ups, mandatory-question gating, live red flags and a 100-point counselling quality audit.
- **Guarantee** plans respect hard constraints: diet pattern (veg / non-veg / vegan / eggetarian / Jain), allergies (zero tolerance), intolerances, dislikes, doctor instructions, day-of-week religious food rules, and medical conditions.

- Treat the dietitian's strategy as a **professional hypothesis**: the AI independently analyses the complete client profile and supports, modifies, replaces, or pauses it — never blindly obeys, never competes.
- Ground every plan's calories and macros in credible food databases (INDB — ICMR-NIN IFCT-based; USDA SR Legacy) instead of model estimates.
- Produce a premium, client-shareable PDF with per-meal calories/macros, daily totals and a weekly summary.
- Fit the ops workflow: each dietitian's scheduled counselling calls (from the ops Google Sheet) appear in-app with a one-click "Start Counselling" that pre-fills the client and closes the loop on completion.
- Zero data leakage between dietitians; never lose intake data mid-call (continuous autosave).

Non-Goals (v2)

- No client-facing portal or login — the dietitian is the only user; plans reach clients as PDFs.
- No payment/subscription management or CRM features.
- No medical diagnosis — the AI flags risks and adapts plans, but the dietitian reviews every plan and owns escalation.

3. USERS

Primary — the practising dietitian (signs in with a @fitelo.co Google account; onboarding captures employee code and phone for ops mapping). Handles 20–100 active clients, runs multiple counselling calls per day from the Today's Activity schedule, needs to capture details fast mid-conversation, and sends every client a weekly chart.

Secondary — the admin. Read-only visibility across dietitians (clients, plans, activity) plus a manual "Sync sheet now" control.

4. USER JOURNEYS

4.1 First counselling (during the call)

1. Dietitian signs in with Google and opens **Today's Activity** — their scheduled calls for the day, synced nightly from the ops sheet — and clicks **Start Counselling** on a row (client code and name pre-fill), or starts a blank + **New Counselling**.
2. Runs the LeanR Premium counselling conversationally, mainly selecting options: goal & deeper motivation, body history, medical safety, digestion, the actual food day (interactive meal timeline), preferences & feasibility, training/protein/recovery, behaviour, lifestyle, dropout risk & coaching profile. Red flags and the quality score update live; everything autosaves.
3. Records the professional assessment (Section 11 — limiting factors, targets, timeline, energy/protein/carb/fat strategy, diet structure, lifestyle-fit score) and the client strategy discussion (Section 12).
4. Clicks **Generate Week 1 Diet Plan**. The AI first runs its **independent clinical review** — on a Pause decision (missing information / safety concern) nothing is saved, the reasons are shown, and the counselling stays a draft to fix and resubmit. Otherwise the plan is generated honouring the review,

grounded against the food database, rendered to PDF and stored; the person becomes a saved client and the appointment is marked Completed.

4.2 Weekly follow-up

1. Dietitian opens the client page, records weight, adherence, energy, hunger, complaints and notes.
2. Clicks **Generate Week N Plan** — the AI follows the LeanR weekly loop (Previous Strategy → Execution → Feedback → Result → Interpretation → Updated Strategy → Next Weekly Diet), sees last week's menu for variety, and produces the adjusted week with a fresh PDF in the plan history.

5. FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT	NOTES
FR-1	Google Workspace sign-in restricted to @fitelo.co ; onboarding collects employee code + phone; profiles carry an admin role for the read-only admin module.	Supabase Auth (Google OAuth), migration 0004
FR-2	Dashboard lists the dietitian's own clients as cards; Today's Activity tab lists their scheduled counselling calls (matched by employee code) with sheet status and a Start Counselling action.	RLS-scoped; QStash daily sync
FR-3	The first counselling is the LeanR Premium bank (Q1–Q105, 12 sections) defined as data: conditional follow-ups, "select up to N" caps where click order = rank, sliders, dates, and the Section 5 interactive meal timeline (time / food+quantity / preparation / source / added components per selected occasion).	Schema-driven; UI renders the schema
FR-4	Continuous autosave (debounced ≤ 2 s) with restore on reload; the draft is cleared only after successful plan generation. Drafts started from an appointment restore only for that appointment.	form_drafts table
FR-5	Mandatory-question gate: a plan can only be generated once every required question (per section, visibility-aware) is answered — enforced in the UI and again server-side.	REQUIRED_IDS
FR-6	Live clinical red flags (urgent symptoms, severe allergy, disordered-eating patterns, pregnancy, renal disease, under-fuelling/RED-S...) surface during the call, distinguish escalations from cautions, and are passed to the AI as hard constraints.	assessment.ts rules
FR-7	Counselling quality audit — a live 100-point score across 11 categories showing exactly what is still missing.	Doubles as progress
FR-8	AI independent clinical review before every Week-1 plan: the AI analyses the complete profile against the dietitian's hypothesis and selects exactly one decision — Support / Support With Minor Modification / Significantly Modify / AI-Led Alternative / Pause . A Pause returns the missing information, persists nothing, and keeps the counselling as a draft. The decision is stored with the plan and shown on the client page.	diet_plans.ai_review (migration 0006)

FR-9	On generation, a strict-JSON 7-day plan honours all restrictions with per-meal calories/protein/carbs/fat; forbidden foods are additionally enforced in code — violations are fed back for correction, surviving dislikes are stripped, and a surviving allergen causes the plan to be refused rather than shipped.	NVIDIA NIM + Zod + deterministic checks
FR-10	Day-of-week food rules (e.g. no non-veg/eggs on Tuesdays & Thursdays) map plan days to real weekdays and are enforced in both the prompt and code.	q38a–c
FR-11	Nutrition grounding: after generation, every meal's macros are recomputed from the INDB + USDA foods table (fuzzy match, household-measure gram weights); a meal keeps model estimates only when not all items match. Grounding failure never blocks a plan.	match_foods_batch RPC, pg_trgm
FR-12	Every plan renders as a branded PDF (per-day meal tables, daily totals, weekly summary, disclaimer) stored in private storage, downloadable via expiring signed URLs.	@react-pdf/renderer
FR-13	Generating the first plan converts the person into a saved client, retains the full counselling in the intake, and marks the source appointment Completed with a link to the client.	clients.intake JSONB
FR-14	Weekly follow-up (weight, adherence, energy, hunger, complaints, notes) generates the next week's plan and keeps client weight current.	followups table
FR-15	If generation fails after the client is saved, the intake is preserved and one-click retry (re-running the AI review) is offered on the client page.	regenerate endpoint
FR-16	Ops sheet sync: the Raw_Counselling Google Sheet is upserted daily into the appointments table via a secret-protected endpoint (QStash schedule); workflow columns survive re-syncs; admins can trigger a manual sync.	/api/sync-counselling

6. NON-FUNCTIONAL REQUIREMENTS

ID	REQUIREMENT
NFR-1	Isolation: Postgres Row Level Security on every table (<code>dietitian_id = auth.uid()</code>); appointments matched by employee code) and folder-scoped storage policies; one dietitian can never read another's data, even with the public API key. Admin access is read-only via policy.
NFR-2	Clinical safety: an allergen surviving all corrective retries aborts the plan; escalated red flags are always visible before generation; a doctor's instruction overrides plan design; Pause decisions block generation until resolved.
NFR-3	Secrets: the NVIDIA API key, service-role key (scripts/sync only) and sheet credentials are server-side only.
NFR-4	Performance: review + generation typically ≤ 120 s; hard cap 300 s with explicit failure and retry path.

NFR-5	Resilience: per-call AI timeouts, automatic fallback model, constrained JSON decoding, schema validation with corrective retries, split (two-call) generation, non-fatal grounding, and a review failure degrading to generation-without-review rather than blocking the dietitian.
NFR-6	Data safety: no intake loss on network failure, generation failure, Pause, or tab close (autosaved drafts + preserved clients).
NFR-7	Deployability: single-command Vercel deploy; database provisioned by sequential SQL migrations (0001–0006); food data seeded by one script.

7. SUCCESS METRICS

- Median time from "counselling complete" to "PDF in hand" $\leq$ 2.5 minutes (review + generation + grounding + render).
- $\geq$ 95% of generation attempts produce a valid plan (including automatic retries/fallback).
- 100% of shipped plans free of client allergens (enforced — refusal instead of shipping).
- $\geq$ 80% of meals grounded in the reference food database.
- Zero cross-tenant data access incidents; zero reported intake-data losses during live calls.

8. FUTURE SCOPE (V3 CANDIDATES)

- **Blood-report upload** → **AI extraction** → **dietitian verification** (the flow the LeanR Premium spec prefers for Q20).
- **Retrieval into the prompt (RAG):** feed candidate foods from the reference DB (filtered by cuisine, diet pattern, restrictions) into generation so every item matches the database.
- WhatsApp/email delivery of the weekly PDF directly to the client.
- Meal-level swap suggestions without regenerating the full week.
- Client progress analytics (weight trend, adherence history) and searchable food selectors in the counselling form.
- Multi-dietitian clinic roles beyond read-only admin.

Review policy: the dietitian's Section 11 strategy is a professional hypothesis; the AI independently tests it and every AI-generated plan is presented to the dietitian before it reaches a client. The PDF footer states the plan was generated with AI assistance and reviewed by the dietitian, and is not a substitute for medical advice.